# ℹ Important information

| | |
|---|---|
| **Course** | Computer systems with project (1DT003) |
| **Examination date** | Thursday 2023-03-16. |
| **Time** | 08:00 - 13:00. |
| **Type** | Digital exam. |
| **Closed book** | This is a closed book exam. This means that no aides such as notes, slides, books, internet searches and collaboration with others are allowed. |

## Instructions

**Part A**

Part A consists of 6 sections, corresponding to module 0-5 in the course, with 10 points each.

To pass the exam, you must:

1. Get at least 3 points in each of the sections in part A
2. Get at least 36 points (60% of max) on part A

Most of the questions are multiple choice-questions where there is only one correct answer.

If you believe there is more than one correct answer, choose the best one based on context and what has been covered in the course.

**Part B**

The problems in part B only needs to be answered if you aim at getting grade 4 or 5 on the exam.

To get grade 4 or 5 on the exam, you have to:

1. First pass part A as described above
2. Out of 20 possible points in part B get...
   ...at least 10 points in part B for an exam grade of 4
   ...at least 15 points in part B for an exam grade of 5

**Questions during exam**

Lars-Åke will visit the exam hall twice during the exam - approximately at 09:00 and 12:00.

**Anonymous exam**

This is an anonymous examination. You will be identified by an anonymous exam code. When the teaching staff marks the exam they will only be able to see the anonymous exam code and not your true identity. Only after the grades have been submitted will your true identity be known.

# 1 Code of honour

I confirm that I will not seek assistance from anyone else to answer the exam questions.

- True
- False

I confirm that I will not use unauthorized resources such as notes, course slides, text books, internet searches etc to answer the exam questions.

- True
- False

---

Totalpoäng: 0

# Comments and assumptions

## Assumptions and comments

If you think that something is wrong with a question or that some information is missing in a question, take a note here.

The questions are numbered. Use the **question number** when you make an assumption or a comment. For example, if you wish to comment on question 7, write **Question 7** on a single line followed by your assumption or comment regarding question 7.

**Write any assumption and/or comments here.**

**Discuss with the teacher when he visits the exam hall!**

- If you get answers from the teacher to any of your questions/assumptions, please remove these questions/assumptions from the above.
- Only leave questions/assumptions you have not been able to discuss with the teacher.

## Part A

# Part A

**...in which you show that you shall pass by:**

- **Getting at least 3 points in each section of part A**
- **Getting at least 36 points in total on part A**

# Mixed concepts

In the table below you find 20 statements. Each statement is identified by a label (A-T).

| | | | |
|---|---|---|---|
| **A** | Cannot be directly accessed from user space. | **K** | Abstraction used to represent the endpoint of network communication |
| **B** | Here are global variablers stored. | **L** | Dynamically considers every request and decides whether it is safe to grant it at this point. |
| **C** | Contains information about the network topology. | **M** | The requirement of ensuring that no two concurrent processes or threads are in the same critical section at the same time. |
| **D** | Communication infrastructure based on statistical multiplexing | **N** | Stores all local variables. |
| **E** | A simplex FIFO channel | **O** | Error-control mechanisum in communication that uses acknowledgments. |
| **F** | Appears to the rest of the system to occur at once without being interrupted. | **P** | When tasks literally run at the same time, e.g., on physical separate cores in a multicore processor. |
| **G** | Suffers from the convoy effect. | **Q** | Are caused by internal events. |
| **H** | Restricts the set of instructions that are allowed to be executed by the CPU. | **R** | Used to request a kernel-level service from the operating system. |
| **I** | Does not suffer from external fragmentation. | **S** | Ensures that at least one of the four necessary conditions for deadlock can never hold. |
| **J** | Smallest sequence of instructions that can be managed independently by a scheduler. | **T** | Used to examine the status of file descriptors of open I/O channels |

## 2 Mixed concepts

In the table to the left you find 20 statements. Each statement is identified by a label (A-T).

Pair each of the 10 concepts below with the statement (A -T) that best describes the concept or best relates to the concept. For each concept, answer by entering the chosen statement label (A - T). There are 10 concepts and 20 statements, hence only 10 of the 20 statements will be among the valid pairings.

| Concept | Statement |
|---|---|
| ARQ | |
| Internet | |
| Mutual exclusion | |
| User mode | |
| Socket | |
| System calls | |
| Deadlock avoidance | |
| Routing table | |
| The data segment | |
| The process control block | |

**Marking:** Each correct pairing will result in **1 point**.

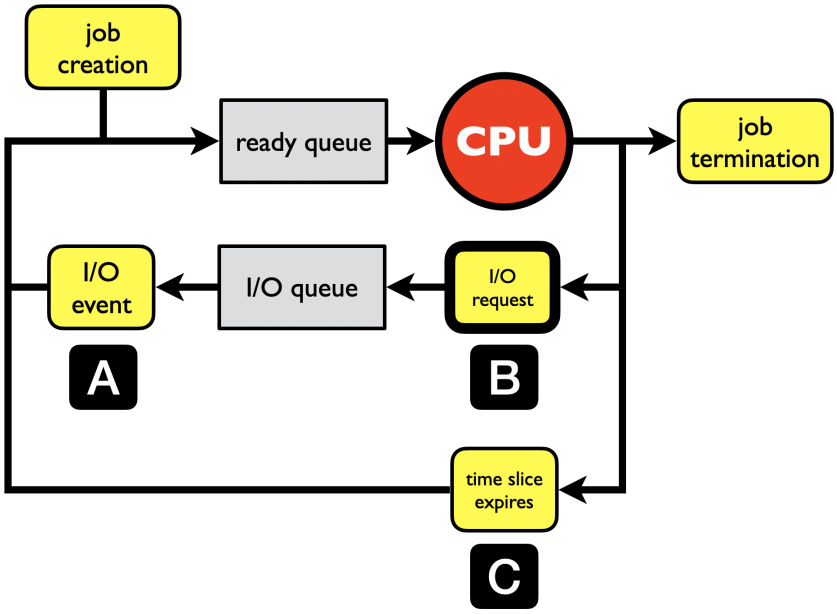
Totalpoäng: 10

# Module 1

Fundamental concepts

## 3 Type of events

The below diagram shows how a process are affected by a number of events.




**Select the correct type of event for A, B and C.**

| | System call | Interrupt | Signal | Random | Static |
|---|---|---|---|---|---|
| A | ○ | ○ | ○ | ○ | ○ |
| B | ○ | ○ | ○ | ○ | ○ |
| C | ○ | ○ | ○ | ○ | ○ |

**Note:** Depending on your screen size and resolution you may need to scroll sideways to see all options.

**Marking:** You get 0.5 point for each correct answer for a maximal total of 2 points.

Totalpoäng: 2

## 4 Memory usage

Below you find a small C-program that compiles without errors.

```c
#include <stdlib.h>
#include <stdio.h>

int A[2];
int *B;

int triple(int x) {
  int z = x + x + x;
  return z;
}

int main(void) {
  B = malloc(2*sizeof(int));
  A[0] = triple(2);
  B[1] = triple(3);
  printf("3*2 = %d\n", A[0]);
  printf("3*3 = %d\n", B[1]);
}
```

**When executing the program, where will the values A[0], B[1], z and the return address when calling triple be stored?**

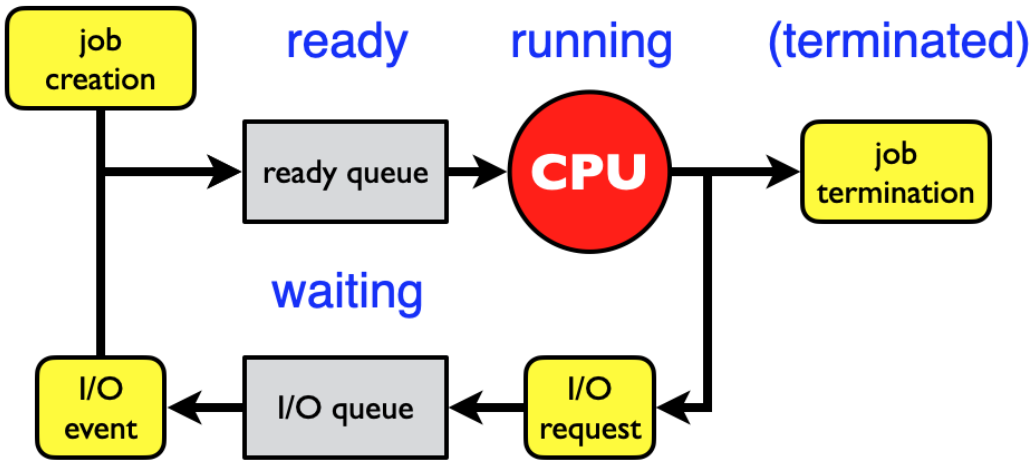
| | Stack segment | Heap segment | Text segment | Data segment |
|---|---|---|---|---|
| The return address when calling triple | ○ | ○ | ○ | ○ |
| B[1] | ○ | ○ | ○ | ○ |
| z | ○ | ○ | ○ | ○ |
| A[0] | ○ | ○ | ○ | ○ |

**Marking:** You get 0.5 point for each correct answer for a maximal 2 points.

Totalpoäng: 2

## 5 Concurrent execution

What model for concurrent execution is described by the following diagram?




- Single-tasking
- Dual-tasking
- Multi-tasking
- Multi-programming

Totalpoäng: 1

## 6 Synchronous and asynchronous events

**From the perspective of an executing process, which of the below events are synchronous and which are asynchronous?**

| | Asynchronous | Synchronous |
|---|---|---|
| Dara read from secondary storage becomes ready | ○ | ○ |
| Making a system call | ○ | ○ |
| Division by zero | ○ | ○ |
| Timer expire | ○ | ○ |

**Marking:** You get 0.5 points for each correct answer.

Totalpoäng: 2

## 7 Internet protocol stack

**The Internet protocol stack**

The picture below illustrates the four layers in the Internet architecture protocol stack. In the left column, specify the name of each layer. In the right column, specify what is sent at each layer from in the case of an application that uses TCP. Answer by dragging the right term to each box.

Hjälp

| | | | |
|---|---|---|---|
| Uppermost layer | | sends: | |
| | | sends: | |
| | | sends: | |
| Lowermost layer | | sends: | |

| | | | | |
|---|---|---|---|---|
| Segment | Link | Transmission | Client | Presentation |
| Control | Socket | Frame | Session | Network |
| Request | Application | Routing | Transport | Message |
| Physical | Packet | Transaction | | |

Totalpoäng: 2

## 8 Sliding window

**Why do we want to use a sliding window at the sender side?**

- To support varying delays in the network
- To support pipelined transmissions with selective repeat
- To support reordering of datagrams in the network
- To support usage of cumulative acknowledgments

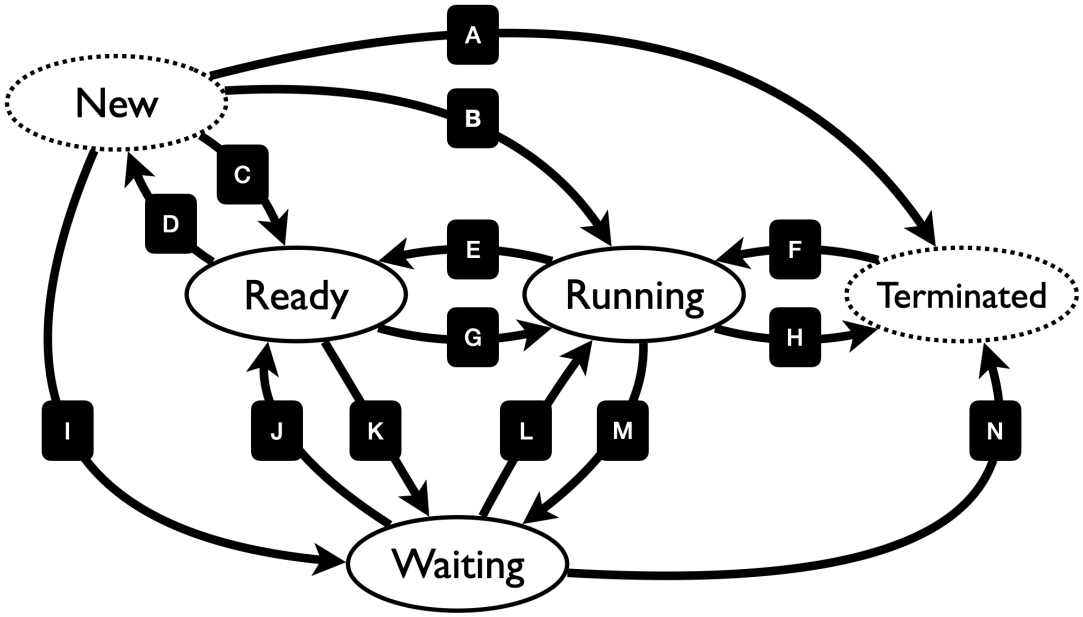
Totalpoäng: 1

# Module 2

The process concept

## 9 State transitions

Processes can transition between the states New, Ready, Running, Waiting and Terminated. In the below diagram, 14 suggested transitions (labeled A, B, C, ..., N ) between these states are shown.




Are all the state transitions shown in the above diagram valid (possible)?

Välj alternativ (Yes, No).

For the three events in the below table, select what state transition is related to this event.

| Event | Transition |
|---|---|
| Time slice expires. | Välj alternativ (A, B, C, D, E, F, G, H, I, J, K, L, M, N) |
| Scheduler dispatch. | Välj alternativ (A, B, C, D, E, F, G) |
| Reading from an empty pipe with at least one open write file descriptor. | Välj alternativ (A, B, C, D, E, F, G, H, I, J, K, L, M, N) |

**Grading:** You get 0.5 points for each correct answer.

Totalpoäng: 2

## 10 CPU context

**The CPU context of a single threaded process is stored:**

- only when making a system call.
- in user space.
- in the PCB.
- on secondary storage.

Totalpoäng: 1

## 11 The fork system call

```
int main(void) {

  pid_t pid = fork();

  switch(pid) {
    case -1:
      exit(EXIT_FAILURE);
    case 0:
      while(true);
    default:
     exit(EXIT_SUCCESS);
  }

}
```

**The above C-program results in the creation of?**

- Two child processes
- No child process
- An orphan process.
- A zombie process.

Totalpoäng: 1

## 12 Pipes and file descriptors

**After a parent process successfully calls the pipe() system call:**

- two new file descriptors will be added to the file descriptor tables of the parent and to the file descriptor table of any children.
- two new file descriptors will be added to the file descriptor table of the parent.
- the parent must use the dup2() system call to add the returned file descriptors to its file descriptor table.
- a write descriptor will be added to the parent file descriptor table and a read descriptor will be added to the file descriptor tables of any children.

---

Totalpoäng: 1

## 13 Illegal memory access

**If during execution, an illegal memory address is accessed, what happens first?**

- A signal is sent.
- The process is terminated.
- The default signal handler is executed.
- An exception is triggered.

---

Totalpoäng: 1

## 14 TCP congestion control

**What is an example of a TCP congestion control mechanism?**

- Reducing the congestion window significantly in the case of a timeout
- Reducing the congestion window when the receiver indicates a near-full buffer
- Reducing the congestion window by 50% in the case of three duplicate acknowledgments
- Reducing the congestion window when observing an increase in measured RTTs

Totalpoäng: 1

## 15 TCP/IP port numbers

**What is the main purpose of port numbers in TCP/IP?**

- Service association
- Address translation
- Integrity checking
- Process multiplexing

Totalpoäng: 1

## 16 TCP and UDP

What is true about the differences between TCP and UDP?
**Select one alternative:**

- TCP establishes a connection before transmission, UDP does not
- TCP uses a checksum for integrity check, UDP has no checksum
- TCP is secure, UDP is insecure
- TCP is segment-oriented, UDP is datagram-oriented

Totalpoäng: 1

# 17 Jitter

**Jitter**
**How is jitter defined in the context of communicating applications?**

- The variation in buffer space needed at the receiving end
- The variation in one-way delay between the sender and the receiver
- The variation in frames per second when playing video frames at the receiver
- The variation in bits per second generated by the sender

---

Totalpoäng: 1

# Module 3

Scheduling and Routing

## 18 I/O bound processes

What characterises I/O- bound processes?

**Complete the sentence below by dragging and dropping the options into the gaps.**

Hjälp

less | more | many | short | few | memory | time | long | energy

An I/O-bound process spends _____ _____ doing I/O and is characterised by _____ _____ CPU bursts.

**Marking:** You must get all four gaps correct to get 1 point.

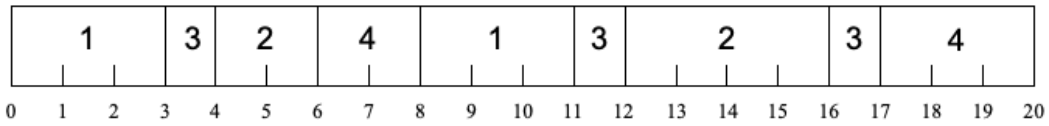
Totalpoäng: 1

## 19 Average response and waiting time

In a CPU scheduling simulation processes arrives to the ready queue according to the following table.

| PID | Arrival | CPU burst |
|---|---|---|
| 1 | 0 | 6 |
| 2 | 3 | 6 |
| 3 | 2 | 3 |
| 4 | 4 | 5 |

The simulation results in the following Gantt chart.




From the above Gantt chart, calculate the average waiting time and the average response time.

**Tip:** Remember that 1/2 = 0.5, 1/4 = 0.25, 3/4 = 0.75.

Average response time: .

Average waiting time:

**Marking:** Each correct answer will result in 1 point for a maximal total of 2 points.

Totalpoäng: 2

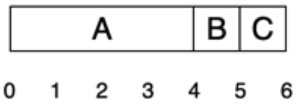
## 20 PSJF

In a CPU scheduling simulation the processes A, B and C arrives to the ready queue according to the following table.

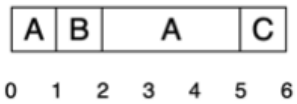
| Pid | Arrival time | Burst |
|---|---|---|
| A | 0 | 4 |
| B | 2 | 1 |
| C | 5 | 1 |

Below you find four Gantt charts for four different schedules of the above processes.

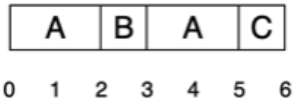
**1**



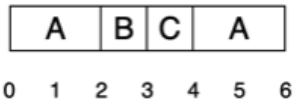

**2**




**3**




**4**




**Which of the above Gantt charts show the PSJF schedule?**

- Gantt chart 1
- Gantt chart 2
- Gantt chart 3
- Gantt chart 4

Totalpoäng: 1

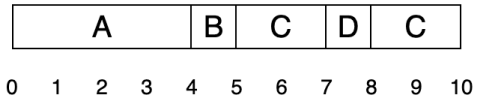
## 21 RR, q = 2

In a CPU scheduling simulation the processes A, B, C and D arrives to the ready queue according to the following table.

| Pid | Arrival time | Burst |
|---|---|---|
| A | 0 | 4 |
| B | 2 | 1 |
| C | 3 | 4 |
| D | 4 | 1 |

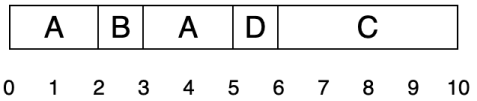
Below you find four Gantt charts for four different schedules of the above processes.

**1**

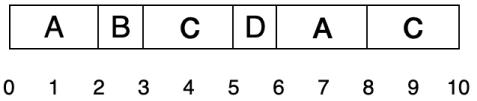



**2**

**3**




**4**




**Which of the above Gantt charts show the RR, q = 2 schedule?**

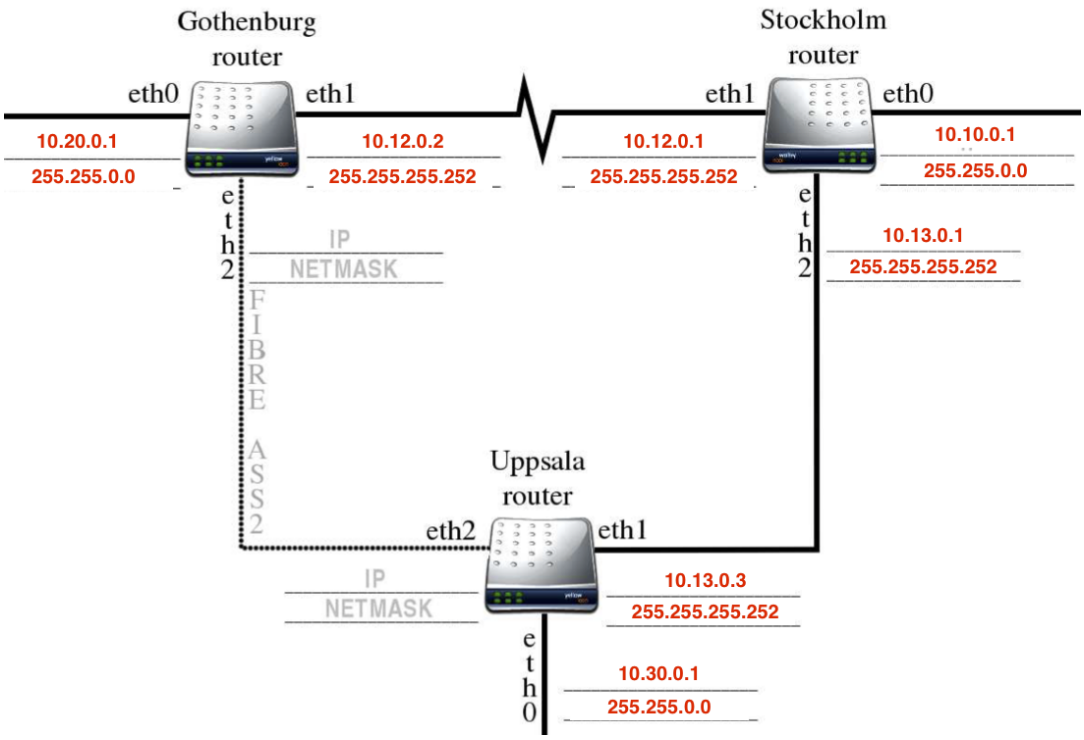
- Gantt chart 1
- Gantt chart 2
- Gantt chart 3
- Gantt chart 4

Totalpoäng: 1

# 22 Routing

**Routing (2p)**

En student sitter och kämpar med routinglaborationen och har nu kommit så långt att hen bestämt sig för en del IP-addresser som skall användas enligt följande figur:




Studenten har börjat skapa en routingtabell (som inte nödvändigtvis är komplett eller helt korrekt) enligt nedan. Dra de rätta värdena enligt studentens plan i ovanstående figur till de tomma luckorna i routingtabellen.

Hjälp

| | | |
|---|---|---|
| eth2 | 10.20.0.0 | 10.30.0.0 |
| 10.12.0.1 | 10.12.0.2 | 255.255.255.255 |
| eth0 | 10.13.0.1 | eth1 |
| 10.13.0.3 | 255.255.255.252 | 255.255.255.0 |

| Routing table at Stockholm router | | | |
|---|---|---|---|
| Destination | Gateway | Genmask | Iface |
| 10.10.0.1 | * | 255.255.0.0 | |
| | 10.12.0.2 | 255.255.0.0 | eth1 |
| 10.30.0.0 | | 255.255.0.0 | eth2 |
| 10.12.0.0 | * | | eth1 |

Totalpoäng: 2

## 23 Netmasks

In the previous problem about the Routing lab, one of the assigned addresses will **not** work. Which one and why?

**Välj ett alternativ:**

- 10.10.0.1 since it has a too short netmask
- 10.13.0.3 since it has a too long netmask
- 10.12.0.1 since it has a too long netmask
- 10.20.0.1 since it will have the same network identifier as 10.12.0.2

Totalpoäng: 1

## 24 CIDR

Three computers on the same network have the same netmask/prefix length and the IP addresses:
10.0.128.2
10.0.190.14
10.0.93.12
**What is the maximum length of their shared netmask/prefix length?**

- 15 bits
- 16 bits
- 17 bits
- 18 bits

Totalpoäng: 1

# 25 Network identifiers

In an IPv4 network, the network identifier can be specified using a netmask.

Two computers in the same organisation are configured as follows:

| Computer | IP Address | Netmask |
|---|---|---|
| A | 10.10.7.116 | 255.255.248.0 |
| B | 10.10.3.94 | 255.255.252.0 |

Now, when A and B wants to exchange data, when will they try to send packets directly to the other part and when will they go through their individually configured default gateway?

**Välj ett alternativ:**

- Both A and B will always send packets to their default gateway for further delivery to the other computer.
- A will always send packets directly to B, while B will send packets to its default gateway for further delivery to A.
- Both A and B will always send packets directly to each other.
- B will always send packets directly to A, while A will send packets to its default gatway for further delivery to B

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Totalpoäng: 1

# Module 4

Threads, synchronization and deadlock.
Access networks.

## 26 Thread resources

**Which of the following resources are shared and which resources are not shared between threads within a process?**

| | Shared | Not shared |
|---|---|---|
| Data segment | ○ | ○ |
| Heap segment | ○ | ○ |
| Text segment | ○ | ○ |
| Stack segment | ○ | ○ |

**Marking:** Each correct answers is rewarded by 0.25 points for a maximal total of 1 points.

Totalpoäng: 1

## 27 Thread creation

**The time required to create a new thread in an existing process is:**

- ○ Less than the time required to create a new process.
- ○ Often equal to, but sometimes slower, than the time required to create a new process.
- ○ Equal to the time required to create a new process.
- ○ Greater than the time required to create a new process.

Totalpoäng: 1

# 28 Conditions for deadlock

**Select the four necessary conditions for deadlock**

- [ ] Progress
- [ ] Starvation
- [ ] Mutual exclusion
- [ ] Circular wait
- [ ] Mutual preemption
- [ ] Hold and wait
- [ ] Fairness
- [ ] Circular exclusion
- [ ] Critical section
- [ ] Bounded wait
- [ ] Exclusive wait
- [ ] No preemption

**Marking**: You get 0.25 points for each correct answer.

---

Totalpoäng: 1

## 29 ACID rule

**Transaction properties**
The ACID rule is used to remember four desirable properties of transactions. But what does the letters stand for?

Choose the right choice for each of the letter below.

Välj alternativ (Atomicity, Authentication, Availability, Adaptiveness, Addressability, Abstraction)

Välj alternativ **(Consistency, Concurrency, Correctness, Checkable, Circular, Computable)**

Välj alternativ **(Intermittent, In-order, Identifiable, Interruptable, Interchangable, Isolation)**

Välj alternativ **(Definable, Declarable, Durability, Disk-storage, Deconstructable, Deterministic)**

Totalpoäng: 1

## 30 Serial equivalence

What of the following can be used to achieve serial equivalence?
**Select one alternative:**

- Time synchronization
- Locks and semaphores
- Transactions
- Permanent storage

Totalpoäng: 1

# 31 Rendezvous

In a system with two threads A and B each thread executes a loop N times. For each iteration:

- thread A prints Ai where i = the iteration number 0, 1, 2, ...., N-1.
- thread B prints Bi where i = the iteration number 0, 1, 2, ...., N-1.

We now want to make the two threads have a rendezvous after each iteration, i.e., the two threads A and B should perform their iterations in lockstep. Lockstep means that they both first perform iteration 0, then iteration 1, then iteration 2, etc. For each iteration the order between A and B should not be restricted.

In the below table you find two examples of valid traces with proper rendezvous (Trace 1 and Trace 2) and two examples of invalid traces (Trace 3 and Trace 4) for N = 3:

| Valid traces | | Invalid traces | |
|---|---|---|---|
| Trace 1 | Trace 2 | Trace 3 | Trace 4 |
| A0 | B0 | A0 | B0 |
| B0 | A0 | A1 | A0 |
| B1 | A1 | B0 | A1 |
| A1 | B1 | B1 | A2 |
| B2 | B2 | B2 | B1 |
| A2 | A2 | A2 | B2 |

Below you find C-like pseudo code for an attempted solution to the rendezvous problem.

```
1   sem semA;
2   sem semB;
3
4   main() {
5     semA = newSem( N );
6     semB = newSem( M );
7
8     spawn(threadA);
9     spawn(threadB);
10  }
11
12  threadA() {
13    for (i = 0; i < N; i++) {
14      wait( X );
15      printf("A%d\n", i);
16      signal(semA);
17    }
18  }
19
20  threadB() {
21    for (i = 0; i < N; i++) {
22      wait( Y );
23      printf("B%d\n", i);
24      signal(semB);
25    }
26  }
27
```

To get the above implementation correct:

What value N (row 5) must be used to initialize semA:

What value M (row 6) must be used to initialize semB:

Which semaphore X (row 14) must be waited on by thread A:

Välj alternativ (semB, semA)

Which semaphore Y (row 22) must be waited on by thread B:

Välj alternativ (semA, semB)

**Grading:** Each correct answer will result in 0.5 point.

## 32 Client configuration

Assume that you are turning on a computer that has been turned off for a week. The computer has no cached mappings from FQDN to IP or from IP to MAC addresses stored.

The first thing that happens when the computer has booted is that it executes a batch script that fetches a web page using HTTP from the server **server.uu.se** .

Fill in the blanks in the text below so that it correctly describes the series of actions.

Hjälp

| | |
|---|---|
| Alternativ | TCP |
| MAC address | UDP |
| NAT box | broadcast |
| IPv4 address | network configuration parameters |
| DNS server | ICMP |
| Netmask | HTTP |
| DNS request | ARP |
| DHCP | default gateway |

First, the computer needs to learn its _____ . This is done by sending a _____ message to get its configuration from a _____ server. Next, it needs to learn the IP address of **server.uu.se** so it sends a _____ to the _____ specified in the configuration. As the destination for the request is not on the same network, the packet must be sent to the _____ for further delivery. To do this, the computer must resolve its _____ using ARP. Once it has that information, the request is sent and the computer learns the IP address of **server.uu.se** in a response message. It can now initiate a _____ connection to **server.uu.se**, send a web request using HTTP and receive the requested data.

## 33 Race conditions and data races

**Which of the following statements about data races and race conditions is true?**

- Data races are a subset of the race conditions.
- All data races are caused by race conditions.
- Race conditions and data races are not a subset of one another, neither the necessary, nor the sufficient condition for one another.
- All race conditions are caused by data races.

---

Totalpoäng: 1

# Module 5

Memory management, files and file systems
Internet history and governance.

## 34 Contiguous memory allocation

**The main problem with contiguous memory allocation is:**

- Internal fragmentation.
- Address binding.
- Random access.
- External fragmentation.

---

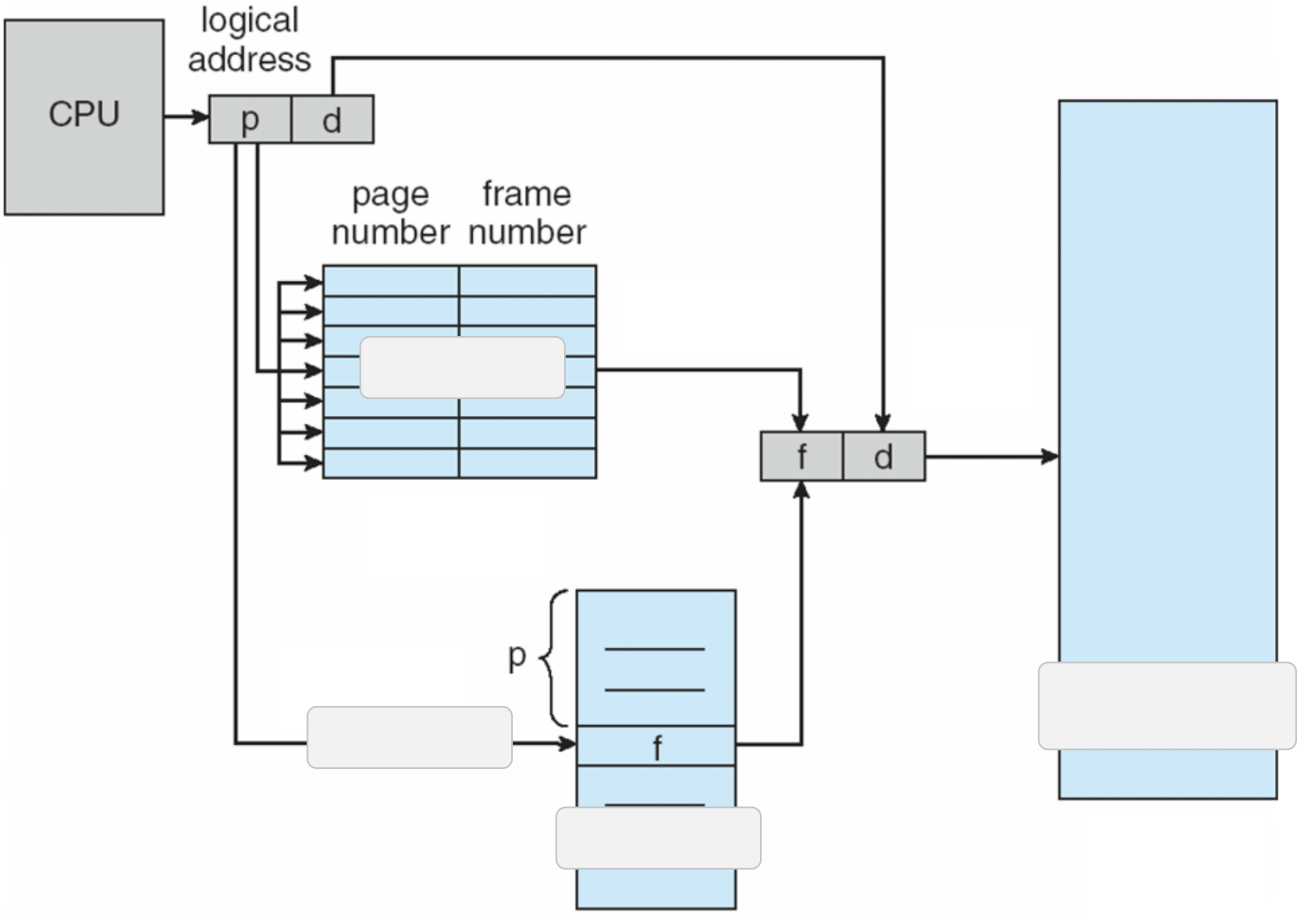
Totalpoäng: 1

## 35 Paging hardware

Drag and drop the correct labels to the four missing labels in the below figure.

Hjälp

| | |
|---|---|
| Page table | Hierarchical page table |
| Virtual memory | TLB hit |
| Data cache | TLB miss |
| MMU | Page table miss |
| TLB | Page table hit |
| Physical memory | Logical memory |




**Marking:** You get 0.25 points for each correct label for a maximum of 1 points.

Totalpoäng: 2

## 36 Disk block allocation

For the three disk block allocation methods below, select the major drawback of each method.

| | No random access | Upper limit on the file size | External fragmention |
|---|---|---|---|
| Linked allocation | ○ | ○ | ○ |
| Contiguos allocation | ○ | ○ | ○ |
| Indexed allocation | ○ | ○ | ○ |

**Marking:** You get 0.25 points for each correct answer and 1 points if you get all three correct.

Totalpoäng: 1

## 37 The Unix iNode

**In a file system using the Unix iNode:**

- There is no internal fragmentation for small files.
- There is only true random access for small files.
- The problem with external fragmentation is only solved for small files.
- There is true random access for all file sizes.

Totalpoäng: 1

## 38 Internet governance

What organisation has the ambition to "*assure the open development, evolution and use of the Internet for the benefit of all people throughout the world*" ?
**Välj ett alternativ:**

- The Internet Society (ISOC)
- The Internet Research Task Force (IRTF)
- The Internet Engineering Task Force (IETF)
- The Internet Governance forum (IGF)

Totalpoäng: 1

## 39 End-to-end principle

Which of the following protocols is a good example of a design that follows the end-to-end principle?
**Välj ett alternativ:**

- ICMP
- DHCP
- TCP
- ARP

Totalpoäng: 1

## 40 Hierarchical page tables

**Hierarchical page tables:**

- Provides faster lookup times than non-hierarchical page tables.
- Takes less space than non-hierarchical page tables.
- Allows for the allocation of many large page tables.
- Eliminates the need for a TLB.

Totalpoäng: 1

## 41 Acyclic graph directories

**In a file system using acyclic graph directories:**

- A file may have multiple absolute paths.
- Directories can only contain soft links to files.
- Directories can only contain hard links to files.
- Does not allows sharing of subdirectories and files between directories.

Totalpoäng: 1

## 42 Internet architecture

The DARPA Internet Architecture had a top-level goal that guided the design of what later became the Internet.
**Which of the following mechanisms that we have talked about in this course relate most to that top-level goal?**

- IP subnetting using CIDR
- TCP congestion control and avoidance
- Node configuration using DHCP
- Best-effort datagram delivery

---

Totalpoäng: 1

## i Part B

# Part B

**...in which you demonstrate yourself worthy to get a higher grade by:**

- **First having passed part A**
- **Scoring at least 10 points (out of 20) for grade "4"**
- **Scoring at least 15 points (out of 20) for grade "5"**

# Scenario för del B

*Denna text beskriver scenariot som frågorna i del B handlar om. Läs igenom texten noga så att du har en tydlig bild av hur scenariot ser ut.*

Du ingår i ett team som ska designa en systemlösning för **samarbetande räddningsrobotar**.

Scenariot är att det har inträffat någon form av naturkatastrof som innebär att byggnader har rasat ihop och infrastruktur för el och mobil datakommunikation har slagits ut. Att söka igenom rasmassorna efter överlevande är ett farligt och samtidigt tidskritiskt arbete.

Rasmassorna består av en labyrintliknande miljö där väggar och golv har rasat ihop på ett sätt som innebär att man inte kan utgå från ritningar över hur byggnaderna en gång såg ut. I vissa sektioner har det skett gasläckage med explosionsrisk vilket innebär att det är områden man vill hålla sig borta från. På olika ställen i rasmassorna finns det överlevande som inte kan ta sig någonstans utan behöver hjälp för att ta sig ut.

Ert system med **samarbetande räddningsrobotar** består av ett antal robotar av olika slag som kan ta sig fram i rasmassorna relativt smidigt. En del av robotarna är markgående, andra svävande. Gemensamt för alla robotar är att de är utrustade med **kamera**, diverse **sensorer** för att kunna upptäcka dels hur omgivningen ser ut (väggar, tak, bråte etc.), men också för att kunna upptäcka **farliga områden med explosionsrisk** samt **överlevande**.

Robotarna är utrustade med **trådlös kommunikation via WiFi.** Eftersom det saknas infrastruktur och det är svårt att täcka in alla skrymslen i rasmassorna via Wifi så kommunicerar robotarna **direkt med varandra** eller med en extern **operatör**. Miljön är dock sådan att den trådlösa kommunikationen när som helst kan brytas för att sedan återkomma senare. Man kan alltså inte räkna med att alla robotar alltid kan kommunicera med varandra eller med den externa operatören.

Robotarnas gemensamma uppdrag är att samarbeta för att kartlägga rasmassorna i jakten på överlevande och producera en **karta över rasmassorna** där de överlevande är tydligt markerade. Kartan måste även tydligt ange **områden med explosionsrisk**. Kartan måste regelbundet uppdateras hos den operatör som övervakar robotarna från utsidan så att man kan skicka in räddningspersonal snarast möjligt efter att en överlevande har återfunnits.

I det här tänkta systemet finns det ett antal aspekter att beakta och det är precis det som frågorna i del B handlar om.

*Kuriosa: I det "gamla" IT-programmet var detta ett projekt som studenterna genomförde under sitt fjärde år på utbildningen. Deras robotar tävlade sedan i en internationell tävling mot andra universitet, oftast med goda resultat.*

# 43 Delad resurs

I det flesta distribuerade datorsystem finns det en eller flera resurser som delas mellan de olika parterna i systemet. I detta fallet är de olika parterna dels varje enskild robot, dels den övervakande operatören.

**Identifiera kritiska delade resurser i detta system** samt vilka **operationer du vill kunna göra mot den delade resursen**. Motivera varför de resurser som du har identifierat är kritiska för att systemet som helhet ska kunna fungera så bra som möjligt.

**Skriv in ditt svar här**

Teckenf...

Ord: 0

Totalpoäng: 4

# 44 Bootstrapping

När robotarna skall aktiveras och skickas in i rasmassor är varje sekund viktig. Därför är det oerhört viktigt att det är enkelt att driftsätta dem. I princip ska man bara kunna slå på dem, ställa dem vid en ingång till rasmassorna, trycka på en knapp och så kör den iväg och påbörjar sitt uppdrag.

**Vilken information behöver roboten få i samband med att den aktiveras för att kunna genomföra sitt uppdrag?** Betänk att alla robotar kanske inte slås på och skickas iväg vid exakt samma tidpunkt. Du kan förutsätta att en robot när den driftssätts har kontakt med den operatör som övervakar robotarnas framsteg i rasmassorna.

**Skriv in ditt svar här**

Teckenf...

Ord: 0

Totalpoäng: 4

# 45 Synkronisering

För att robotarna skall kunna samarbeta så effektivt som möjligt mot det gemensamma målet att kartlägga rasmassorna och hitta överlevande så snabbt som möjligt behöver de ibland synkronisera viss information mellan sig.

**Vad behöver man synkronisera** mellan varandra och ungefär hur skulle man kunna gå tillväga för att uppnå det? Du behöver inte komma med en ingående teknisk redogörelse utan kan beskriva processen skissartat.

**Skriv in ditt svar här**

Teckenf...

Ord: 0

Totalpoäng: 4

# 46 Koordinering

För att kunna genomföra den gemensamma uppgiften så effektivt som möjligt finns även ett behov av **koordination**. Ge exempel på vad som kan behöva koordineras mellan de olika robotarna samt beskriv skissartat hur det kan gå till. Om det underlättar kan även den externa operatören involveras i koordinationsarbetet.

**Skriv in ditt svar här**

Teckenf...

Ord: 0

Totalpoäng: 4

# 47 Felhantering vid kommunikationsbortfall

En inneboende problematik i scenariot är att en utforskande robot när som helst kan t**appa kontakten** med övriga robotar eller med operatören. Detta skall dock inte hindra den från att fortsätta att ta sig fram i rasmassorna och samla in data eftersom den förr eller senare sannolikt återfår kontakten med andra robotar eller med operatören.

Beskriv kortfattat hur du tänker dig att man kan hantera kommunikationsbortfall samt hur man hanterar den situation som uppstår när man återfår kontakten med en annan robot eller med operatören.

**Skriv in ditt svar här**

Teckenf...

Ord: 0

Totalpoäng: 4